

Andrea D'Angelo
dangeloandrea14@gmail.com
andrea-dangelo.com/

Short Bio

Andrea D'Angelo is a Postdoctoral researcher at Aarhus University and, from November 2026, a Danish Data Science Academy (DDSA) Postdoctoral Fellow with his own funded project on *Robust Graph Unlearning*. He was awarded a Ph.D. cum laude with the Doctor Europaeus distinction by the University of L'Aquila in 2026. His research focuses on Trustworthy Artificial Intelligence and Machine Unlearning. He has authored more than 15 papers in leading venues, including NeurIPS, EMNLP, IJCAI, CIKM, and the Machine Learning journal. He has organized international workshops, delivered a tutorial at ECML-PKDD 2025, served as Proceedings Chair for CIKM 2024 and WSDM 2026, and given invited talks at leading institutions, including the Technical University of Munich (TUM).



Research Experience

Nov 2026–	DDSA Postdoc , Datalogi, Aarhus University. Project: <i>Robust Graph Unlearning</i> . Funded by the Danish Data Science Academy.
May–Oct 2026	Postdoctoral Researcher , Datalogi, Aarhus University, (Host: Prof. Davide Mottin).
Dec 2025–Apr 2026	Research Scholarship , University of L'Aquila, industry collaboration with Appendo. Appendo research group member since 2026.
2025	Visiting Ph.D. Student (6 months), Datalogi, Aarhus University. Graph Unlearning and Differential Privacy.
2022–26	Ph.D. Student (XXXVIII cycle), University of L'Aquila.
2021–22	Research Scholarship (9 months), University of L'Aquila, <i>Territori Aperti</i> project. LLM-based Dense Retrieval.

Education

2022–26	Ph.D. , Computer Science, University of L'Aquila, cum Laude, Doctor Europaeus. Thesis: " <i>Trustworthy AI Through Principled Advances in Machine Unlearning, Algorithmic Bias Mitigation, and Explainability</i> "
2020–22	M.Sc. , Computer Science (LM-18), University of L'Aquila, 110/110 cum Laude, 22/10/2022. Thesis: " <i>Clouds of vectors for density-based Information Retrieval</i> ".
2016–19	B.Sc. Computer Science (L-31), University of L'Aquila, 110/110 cum Laude, 21/10/2019. Thesis: " <i>Application of standard compilation techniques to non-linear languages</i> ".

Honours & Awards

2026	Awarded DDSA Postdoc Grant (Danish Data Science Academy) for the project titled <i>Robust Graph Unlearning</i> (1,365,000 DKK / ~€183,000).
2024	Awarded EUGAIN Virtual Mobility Grant , EUGAIN COST Action E-COST-GRANT-CA19122-04419a60.

Teaching

LUISS University of Rome

2026	Lecturer, Big data management: data mining and analysis for management decisions, 21 hours
------	--

Engage.EU Summer School (LUISS University of Rome)

2026	Lecturer, Privacy Preservation for Machine Learning, 2 hours
------	--

University of L'Aquila

2024	Co-Creator of the MOC course on "Reinforcement Learning" of the SoBigData Academy ¹ .
2023, 2024	Assistance During Lectures, Deep Neural Networks (Computer Science Msc, LM-18), 8 hours.
2023, 2024	Assistance During Lectures, Artificial Intelligence (Computer Science Msc, LM-18), 5 hours.

Research Software



Lethe

Lethe is an open-source benchmark for Edge Unlearning in Graph Neural Networks. It provides a standardized suite of link inference attacks for evaluating edge unlearning methods on structurally informative graphs. Accepted at **NeurIPS 2026** (Rank A*), Evaluations and Datasets Track.



ERASURE

ERASURE is an open-source Python framework designed to make Machine Unlearning research reproducible and comparable. It provides a configuration driven interface for running and benchmarking forgetting algorithms across multiple models, datasets, and evaluation metrics. Published as a resource paper at **CIKM 2025** (Rank A) and demonstrated at **IJCAI 2025** (Rank A*).

¹<https://sobigdata.unipi.it/course/view.php?id=17>

Supervision

Scholars

2025	Francesca Ciccarelli, University of L'Aquila, Ph.D. Student, on the topics of Machine Unlearning applied to LLMs.
2025	Manuel Antonio Vilas Valiente, University of L'Aquila, Visiting Scholar, on the topics of Machine Unlearning applied to Time Series.
2024, 2025	Tutor for Pinkamp, a STEM Outreach program.
2026	Outreach talk <i>Opportunities and Privacy Risks with Artificial Intelligence</i> , Istituto Tecnico Marconi, Penne.

Theses

2026	Susanna Peretti, University of Urbino, Computer Science Master's Degree, " <i>Machine Unlearning applied to Time Series and Human Activity Recognition</i> "
2025	Fabio D'Andreamatteo, University of L'Aquila, Data Science Master's Degree, " <i>HybridKAN: Leveraging Multi-Sized Sub-MLPs for Enhanced Performance</i> ".
2025	Samuele Ramassone, University of L'Aquila, Computer Science Master's Degree, " <i>Beyond the Systematic Evaluation of Large Language Models: From Theory to Practical Analysis of Results</i> ".
2025	Gabriele Tagliente, University of L'Aquila, Computer Science Master's Degree, " <i>From ERASURE to Evaluation of Machine Unlearning Techniques</i> ".
2025	Matteo Iezzi, University of L'Aquila, Data Science Master's Degree, " <i>Recent Advances in Chain-of-Thought Prompting: Current Trends and Automation Approaches</i> ".

Service

Organization

2026	Co-organizer of Parallel Session "Knowledge Graphs: Connecting Theory, Algorithms, and Applications", D3A 4.0 (Danish Digitalization, Data Science and AI), Nyborg Strand, Denmark.
2026	Co-organizer of Workshop on Innovations, Privacy-preservation, and Evaluations of machine Unlearning Techniques (WIPE-OUT 2), ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases) 2026, Naples.
2025	Co-organizer of Workshop on Web & Graphs, Responsible Intelligence, and Social Media (WEB&GRAPH 2026), WSDM (The 19th ACM International Conference on Web Search and Data Mining) 2026, Boise, Idaho, USA.

2025	Proceedings Chair of WSDM (ACM International Conference on Web Search and Data Mining) 2026.
2025	Co-organizer of Workshop on Innovations, Privacy-preservation, and Evaluations of machine Unlearning Techniques (WIPE-OUT 2025), ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases) 2025, Porto.
2025	Program Committee Chair of Workshop on Innovations, Privacy-preservation, and Evaluations of machine Unlearning Techniques (WIPE-OUT 2025), ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases) 2025, Porto.
2025	Co-organizer of Tutorial: Unraveling Graph Counterfactual Explainability: from Theoretical Foundations to Technical Mastery, ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases) 2025, Porto.
2024	Proceedings Chair of CIKM (ACM International Conference on Information and Knowledge Management) 2024.
2024	Co-organizer of "Territori Aperti: dai risultati alle sfide future" event.

Member of the Program Committee

2026	Springer Artificial Intelligence Review Journal
2026	Automated Software Engineering Journal
2027	AAAI (AAAI Conference on Artificial Intelligence).
2026	ITADATA (Italian Conference on Big Data and Data Science).
2025	ACM Web Conference (formerly known as the International World Wide Web Conference)
2025, 2026	KDD (ACM SIGKDD Conference on Knowledge Discovery and Data Mining) Datasets & Benchmarks.
2024, 2025, 2026	CIKM (ACM International Conference on Information and Knowledge Management). Distinguished Reviewer in 2024.
2025, 2026	ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases).
2025	ICPE Data Challenge (International Conference on Performance Engineering).
2024, 2026	LL4MSE Workshop @ STAF (Software Technologies: Applications and Foundations).
2024	DELTA Workshop @ KDD 2024 (ACM SIGKDD Conference on Knowledge Discovery and Data Mining).
2024	MIDAS Workshop @ ECML-PKDD 2024 (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases).
2023	BIAS Workshop @ ECIR 2023 (European Conference on Information Retrieval).

Presentations

Invited Talks

- IT1. **D'Angelo, Andrea.** *Perspectives on Machine Unlearning and Privacy-preserving Machine Learning* Technical University of Dresden. Feb. 2026.
- IT2. **D'Angelo, Andrea.** *Trustworthy AI and the Right to be Forgotten: The Role of Machine Unlearning* LUISS Guido Carli. July 2026.
- IT3. **D'Angelo, Andrea.** *Machine Unlearning: Safeguarding Privacy and Security in AI Models* Technical University of Munich. July 2024.
- IT4. **D'Angelo, Andrea.** *Privacy Implications in the Training and Use of Large Language Models* Territori Aperti, Napoli. 2024.
- IT5. **D'Angelo, Andrea.** *Uncovering Gender Bias in Academia* AIGAP 2023 (Algorithmic Biases in Artificial Intelligence from Interdisciplinary Perspectives), L'Aquila. 2023.

Talks

- T1. **D'Angelo, Andrea.** *Forgetting Right: Towards Reproducible Benchmarks in Machine Unlearning with ERASURE* WIPEOUT Workshop @ ECML PKDD 2025 (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases), Porto. 2025.
- T2. **D'Angelo, Andrea.** *HybridKAN: Leveraging Multi-Sized Sub-MLPs for Enhanced Performance* ITADATA (Italian Conference on Big Data and Data Science) 2025, Turin. 2025.
- T3. **D'Angelo, Andrea.** *A Novel Relevance Score for Unsupervised Retrieval with Large Language Models* KDD 2024 (ACM SIGKDD Conference on Knowledge Discovery and Data Mining), Barcelona. 2024.
- T4. **D'Angelo, Andrea.** *Echocardiographic Epicardial Adipose Tissue Quantification: Challenges and Insights* Percom (International Conference on Pervasive Computing and Communications) 2024, Biarritz. 2024.
- T5. **D'Angelo, Andrea.** *Data-Driven Analysis of Gender Fairness in the Software Engineering Academic Landscape* ECSA 2023 (European Conference on Software Architecture), Istanbul. 2023.

Posters

- P1. Ciccarelli, F., **D'Angelo, Andrea** & Stilo, G. *Towards a Novel Visual Evaluation of Algorithmic Bias: Insights on the Italian Academic System* KDD (ACM SIGKDD Conference on Knowledge Discovery and Data Mining), 2024.
- P2. **D'Angelo, Andrea**, Di Marco, A. & Stilo, G. *A novel Relevance Score for Unsupervised Retrieval with Large Language Models* KDD (ACM SIGKDD Conference on Knowledge Discovery and Data Mining) 2024, Barcelona, Spain, 2024.

Tutorials

- D1. Prenkaj, B., Zaradoukas, E., Stilo, G., **D'Angelo, Andrea** & Prado-Romero, M. *Unraveling Graph Counterfactual Explainability: from Theoretical Foundations to Technical Mastery* ECML-PKDD 2025, Porto, Portugal. Sept. 2025.

Publications

G Google Scholar: <https://scholar.google.com/citations?user=9WIwwAAAAJ>

Peer-reviewed Conference Papers

Reported conference ranks are from CORE.

- C1. Bushipaka, P., **D'Angelo, Andrea**, Passaro, L. & Cucinotta, T. *GRACE: Gradient-guided Coreset Selection for LLM Unlearning in Findings of the Association for Computational Linguistics: EMNLP 2026 [A*]* (Association for Computational Linguistics, 2026).
- C2. **D'Angelo, Andrea**, Gullo, F., Mottin, D., Ranu, S. & Stilo, G. *Lethe: Link Inference Attacks For Evaluation of Edge Unlearning Methods in Advances in Neural Information Processing Systems (NeurIPS 2026 [A*]), Evaluations and Datasets Track* (2026).
- C3. D'Andreamatteo, F., **D'Angelo, Andrea** & Stilo, G. *HybridKAN: Leveraging Multi-Sized Sub-MLPs for Enhanced Performance in ITADATA (Italian Conference on Big Data and Data Science) 2025* (Torino, Italy, 2025).
- C4. **D'Angelo, A.**, Savelli, C., Tagliente, G., Giobergia, F., Baralis, E. & Stilo, G. *How to Make Reproducible Research in Machine Unlearning with ERASURE in Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25 [A*] Demo Track* (International Joint Conferences on Artificial Intelligence Organization, Aug. 2025), 11025–11029. <https://doi.org/10.24963/ijcai.2025/1255>.
- C5. **D'Angelo, Andrea**, Savelli, C., Tagliente, G., Giobergia, F., Baralis, E. & Stilo, G. *ERASURE: a Modular and Extensible Framework for Machine Unlearning in Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM 2025 [A])* (Association for Computing Machinery, Seoul, Korea, 2025).
- C6. Di Sipio, C., **D'Angelo, Andrea**, Rubei, R. & Politowski, C. *On the Need for Reproducibility Guidelines for Open-Source Games: A itch.io Case Study in 2025 IEEE Conference on Games (CoG) [C]* (2025), 1–8.
- C7. d'Aloisio, G., **D'Angelo, Andrea**, Marzi, F., Di Marco, D., Stilo, G. & Di Marco, A. *Data-Driven Analysis of Gender Fairness in the Software Engineering Academic Landscape in Software Architecture. ECSA 2023 Tracks, Workshops, and Doctoral Symposium* (eds Tekinerdoğan, B., Spalazzese, R., Sözer, H., Bonfanti, S. & Weyns, D.) (Springer Nature Switzerland, Cham, 2024), 89–103. ISBN: 978-3-031-66326-0.
- C8. **D'Angelo, Andrea** & d'Aloisio, G. *Grammar-Based Anomaly Detection of Microservice Systems Execution Traces in Companion of the 15th ACM/SPEC International Conference on Performance Engineering* (Association for Computing Machinery, London, United Kingdom, 2024), 77–81. ISBN: 9798400704451. <https://doi.org/10.1145/3629527.3651844>. Best Short Paper Award (ICPE 2024).
- C9. **D'Angelo, Andrea**, Di Sipio, C., Politowski, C. & Rubei, R. *PlayMyData: a curated dataset of multi-platform video games in Proceedings of the 21st International Conference on Mining Software Repositories [A]* (Association for Computing Machinery, Lisbon, Portugal, 2024), 525–529. ISBN: 9798400705878. <https://doi.org/10.1145/3643991.3644869>.
- C10. Bianchi, A., d'Aloisio, G., **Andrea D'Angelo**, Marco, A. D., Matteo, A. D., Leone, J., Scoccia, G., Stilo, G. & Traini, L. *DIORAMA: Digital twIn fOR sustAinable territorial MANagement in Proceedings of itaDATA (Italian Conference on Big Data and Data Science) (Sept. 2022)*, 144–155.

- C11. d'Aloisio, G., Stilo, G., Di Marco, A. & **D'Angelo, Andrea**. *Enhancing Fairness in Classification Tasks with Multiple Variables: A Data- and Model-Agnostic Approach* in *Advances in Bias and Fairness in Information Retrieval* (eds Boratto, L., Faralli, S., Marras, M. & Stilo, G.) (Springer International Publishing, Cham, 2022), 117–129. ISBN: 978-3-031-09316-6.

Journal Articles

Reported journal ranks are from Scimago Journal rank (SJR).

- J1. D'Angelo, A., Savelli, C., Giobergia, F., Baralis, E. & Stilo, G. On the Evaluation of Machine Unlearning Methods: A Multi-domain Classification Benchmark. *Machine Learning [Q1]* **115**, 161. <https://doi.org/10.1007/s10994-026-07094-y> (2026).
- J2. D'Angelo, A., Gullo, F. & Stilo, G. The Forget-Set Identification Problem. *Machine Learning [Q1]* **114**. <https://doi.org/10.1007/s10994-025-06897-9> (2025).
- J3. **Andrea D'Angelo**, d'Aloisio, G., Marzi, F., Di Marco, A. & Stilo, G. Uncovering gender gap in academia: A comprehensive analysis within the software engineering community. *Journal of Systems and Software [Q1]* **217**, 112162. ISSN: 0164-1212. <https://www.sciencedirect.com/science/article/pii/S0164121224002073> (2024).
- J4. Bianchi, A., Zelli, V., **D'Angelo, Andrea**, Di Matteo, A., Scoccia, G., Cannita, K., Dimas, A. S., Glentis, S., Zazzeroni, F., Alesse, E., Di Marco, A. & Tessitore, A. A method to comprehensively identify germline SNVs, INDELs and CNVs from whole exome sequencing data of BRCA1/2 negative breast cancer patients. *NAR Genomics and Bioinformatics [Q1]* **6**, lqae033. ISSN: 2631-9268. eprint: <https://academic.oup.com/nargab/article-pdf/6/2/lqae033/57258946/lqae033.pdf>. <https://doi.org/10.1093/nargab/lqae033> (Apr. 2024).
- J5. d'Aloisio, G., **Andrea D'Angelo**, Di Marco, A. & Stilo, G. Debiasser for Multiple Variables to enhance fairness in classification tasks. *Information Processing & Management [Q1]* **60**, 103226. ISSN: 0306-4573 (2023).
- J6. Della Penna, G., Orefice, S. & **D'Angelo, A.** Exploiting spatial relations for grammar-based specification of multidimensional languages. *Knowledge and Information Systems [Q1]* **65**, 3995–4020 (2023).

Book Chapters

- B1. Takaoka, A. J. W., Alboaie, L., Cutrupi, C. M., **Andrea D'Angelo**, Marco, A. D. & Hillston, J. *Best Practices and Lessons from Academia-Industry Collaboration Initiatives* Mar. 2025.

Other Grants & Recognitions

2025	Awarded ERASMUS Traineeship Grant for European mobility.
2024	Awarded CIKM (ACM International Conference on Information and Knowledge Management) 2024 Distinguished Reviewer Award .
2024	Awarded Mobility Grant from KDD 2024 (ACM SIGKDD Conference on Knowledge Discovery and Data Mining).
2024	Awarded Best Short Paper Award at ICPE 2024 (International Conference on Performance Engineering).

2022

Awarded publicly funded Ph.D. Scholarship at University of L'Aquila.

Last updated: September 29, 2026